Chapter 9: Cellular Respiration and Fermentation

Comprehensive Enhanced Master Study Guide • Bioenergetics, Catabolic Pathways & Chemiosmosis

Course: BSC2010C General Biology I **Institution:** University of Central Florida **Format:** Fused Modular Master Syllabus

Module 1: Principles of Redox Chemistry & Catabolic Electron Harvest

Redox Mechanics: Oxidation and Reduction

Cellular respiration is a series of coupled catabolic oxidation-reduction (redox) reactions that harvest energy stored in the chemical bonds of glucose to synthesize ATP:

- **Oxidation (LEO):** Loss of electrons from a substance.
- **Reduction (GER):** Gain of electrons by a substance.
- **Mnemonic:** "LEO the lion says GER" • *Lose Electrons Oxidation; Gain Electrons Reduction.*



Figure 9.1: The LEO-GER Mnemonic. *Foundational memory anchor for biological oxidation and reduction.*

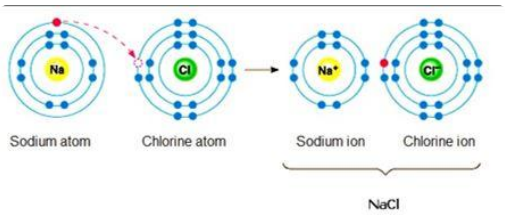


Figure 9.2: Ionic Redox Model. *Sodium loses an electron (oxidized to Na^+); Chlorine gains an electron (reduced to Cl^-).*

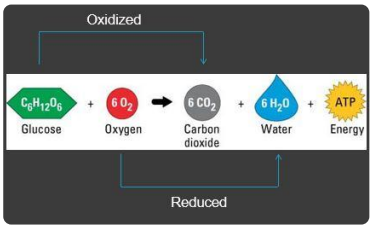


Figure 9.3: Cellular Respiration as an Exergonic Redox Process. *Glucose is oxidized to CO_2; Oxygen is reduced to H_2O (ΔG = -686 kcal/mol).*

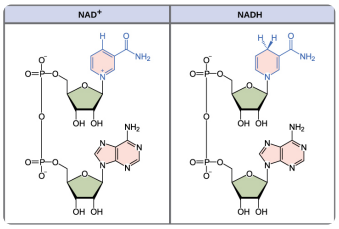


Figure 9.4: NAD^+ / NADH Dinucleotide Architecture. *Nicotinamide accepts 2 e^- and 1 H^+ to form high-energy NADH.*

Coenzymes: The Mobile Electron Shuttles

Electrons are stripped from fuels by dehydrogenases and transferred with protons to mobile coenzyme shuttles:

- **NAD^+ → NADH:** Accepts 2 electrons and 1 proton (H^+), releasing a second H^+ into the matrix. Each NADH carries substantial reducing power to feed Complex I of the ETC.
- **FAD → $FADH_2$:** Accepts 2 electrons and 2 protons. Enters downstream at Complex II.

Module 2: Phosphorylation Mechanisms • Substrate-Level vs. Oxidative

Mechanisms of ATP Generation

Cellular respiration generates ATP via two distinct biochemical mechanisms across different compartments:

- **Substrate-Level Phosphorylation:** Direct enzymatic transfer of a phosphate group from a phosphorylated metabolic intermediate to ADP by a kinase enzyme. Occurs during Glycolysis (cytosol) and the Citric Acid Cycle (matrix). Produces a minor portion (~4 ATP) of total cellular ATP.
- **Oxidative Phosphorylation:** Chemiosmotic ATP synthesis powered by redox reactions of the electron transport chain. Inner membrane-bound ATP synthase harnesses a proton gradient to phosphorylate ADP with inorganic phosphate (P_i). Accounts for ~90% of aerobic ATP (~26–28 ATP).

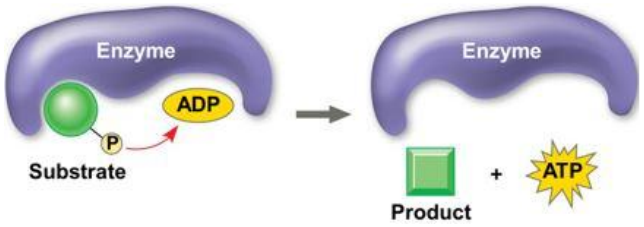


Figure 9.5: Substrate-Level Phosphorylation. *A soluble kinase enzyme couples exergonic phosphate release from a high-energy substrate directly to ADP synthesis.*

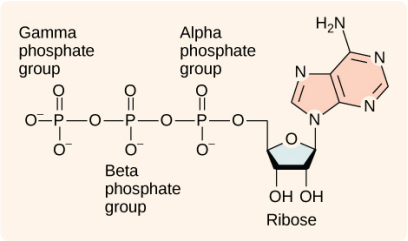


Figure 9.6: ATP Triphosphate Tail Architecture. *Alpha, beta, and gamma phosphates with high-energy phosphoanhydride bonds.*

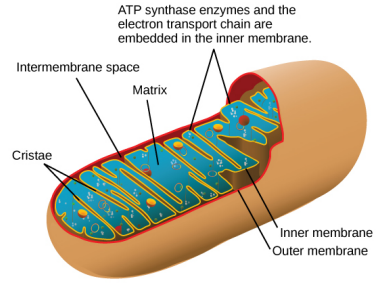


Figure 9.7: Mitochondrion Compartmentalization. *Outer membrane, cristae-folded inner membrane, intermembrane space (H^+ reservoir), and matrix (enzymes).*

Comparison Matrix: Phosphorylation Mechanisms

Parameter	Substrate-Level	Oxidative
Enzyme Type	Soluble Kinase	CF_0CF_1 ATP Synthase
Phosphate Source	Phosphorylated organic substrate	Free inorganic phosphate (P_i)
Driving Force	Exergonic substrate cleavage	Proton-motive force (chemiosmosis)
Compartments	Cytosol & Mitochondrial Matrix	Inner Mitochondrial Membrane
Aerobic Dependence	Occurs with or without O_2	Strictly dependent on terminal O_2

Module 3: Stage 1 • Glycolysis (The Cytosolic Splitting of Glucose)

Glycolytic Overview & Evolutionary Antiquity

Glycolysis ("sugar splitting") is a 10-step metabolic pathway in the cytosol that degrades 1 six-carbon glucose into 2 three-carbon pyruvates without utilizing molecular oxygen:

- **Location:** Cytoplasm / Cytosol of all living cells.
- **Oxygen Independence:** Operates anaerobically as well as aerobically; evolved in ancient anaerobic prokaryotes before Earth's atmosphere accumulated O_2.

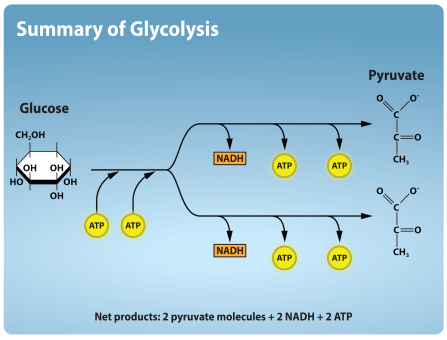


Figure 9.8: The Two Phases of Glycolysis. *Energy Investment Phase consumes 2 ATP; Energy Payoff Phase generates 4 ATP and 2 NADH.*

Net Stoichiometry per Glucose Molecule

Glucose + 2 NAD^+ + 2 ADP + 2 P_i → 2 Pyruvate + 2 NADH + 2 H^+ + 2 ATP (net)

Phase 1: Energy Investment Phase (Steps 1–5)

The cell invests **2 ATP molecules** to phosphorylate and destabilize the hexose ring:

- **Step 1 (Hexokinase):** Transfers phosphate from ATP to glucose, forming glucose-6-phosphate. Traps glucose inside the cell.
- **Step 3 (Phosphofructokinase - PFK):** Phosphorylates fructose-6-phosphate to fructose-1,6-bisphosphate using a second ATP. *PFK is the major allosteric pacemaker of cellular respiration* (inhibited by ATP and citrate; activated by AMP).
- **Step 4–5:** Fructose-1,6-bisphosphate is cleaved into two triose isomers: Dihydroxyacetone phosphate (DHAP) and **Glyceraldehyde-3-phosphate (G3P)**. Isomerase converts all DHAP into G3P (2 G3P per glucose).

Phase 2: Energy Payoff Phase (Steps 6–10)

Energy released from fuel oxidation is harvested to synthesize ATP and NADH (multiplied by 2 per glucose):

- **Step 6 (Dehydrogenase):** G3P is oxidized and phosphorylated to 1,3-bisphosphoglycerate, reducing **2 NAD^+ to 2 NADH**.
- **Step 7 (Phosphoglycerate Kinase):** Direct transfer of phosphate to ADP produces **2 ATP** via substrate-level phosphorylation.
- **Step 10 (Pyruvate Kinase):** Phosphoenolpyruvate (PEP) transfers its phosphate to ADP, producing **2 ATP** and yielding **2 Pyruvate**.

Module 4: Stage 2 • Pyruvate Oxidation (The Mitochondrial Entryway)

Mitochondrial Import & Compartmental Transition

In the presence of O_2, pyruvate produced in the cytosol is actively transported across the outer and inner mitochondrial membranes into the **mitochondrial matrix** via a dedicated **pyruvate translocase carrier protein**.

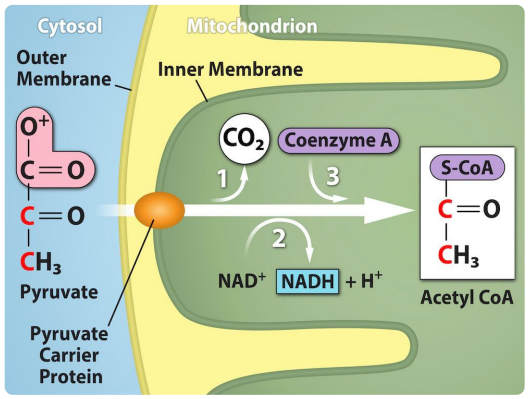


Figure 9.9: Three-Step Pyruvate Dehydrogenase Mechanism.
Decarboxylation (-CO_2), oxidation to acetate (reducing NAD^+ to NADH), and attachment of Coenzyme A forming Acetyl-CoA.

The Pyruvate Dehydrogenase Multienzyme Complex

Once inside the matrix, a massive multienzyme assembly (pyruvate dehydrogenase) catalyzes three consecutive steps:

1. **Decarboxylation:** Pyruvate's carboxyl group ($-COO^-$) is cleaved and released as a molecule of **CO_2** (the first carbon waste of respiration).
2. **Oxidation:** The remaining two-carbon acetyl fragment is oxidized to acetate. The extracted electrons are transferred to NAD^+, reducing it to **NADH + H^+**.
3. **Coenzyme A Attachment:** An unstable, high-energy sulfur bond connects the acetyl group to Coenzyme A (derived from pantothenic acid / vitamin B5), forming **Acetyl-CoA**.

Stoichiometric Yield per Glucose (2 Pyruvates)

2 Pyruvate + 2 NAD^+ + 2 CoA → 2 Acetyl-CoA + 2 NADH + 2 H^+ + 2 CO_2

- **2 Acetyl-CoA:** Enter the Citric Acid Cycle in the matrix.
- **2 NADH:** Donate electrons to Complex I of the ETC.
- **2 CO_2:** Diffuse out of the mitochondrion and cell; exhaled.

Clinical Toxin Target: Arsenic Poisoning

Arsenic inhibits the pyruvate dehydrogenase complex. Pyruvate cannot be converted to Acetyl-CoA, causing massive pyruvate accumulation while starving downstream respiration of fuel, drastically reducing ATP synthesis.

Module 5: Stage 3 • The Citric Acid Cycle (Krebs Cycle / TCA Cycle)

Overview: Complete Carbon Oxidation in the Matrix

The **Citric Acid Cycle** is an 8-step metabolic loop located in the mitochondrial matrix that completes the oxidation of organic fuel molecules derived from glucose:

- **Entry:** Two-carbon Acetyl-CoA condenses with four-carbon **oxaloacetate** to form six-carbon **citrate**.
- **Regeneration:** The cycle oxidizes two carbons (released as 2 CO_2) and regenerates oxaloacetate for another turn.

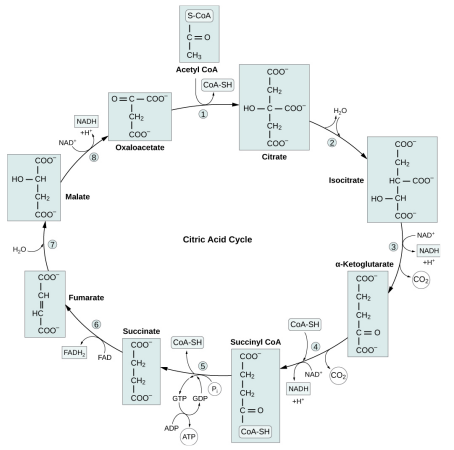


Figure 9.10: The 8 Steps of the Citric Acid Cycle. *Acetyl-CoA enters; 2 CO_2 released; 3 NADH, 1 $FADH_2$, and 1 ATP generated per turn.*

Step-by-Step Energy Harvesting Accounting

Reaction Step	Key Enzyme	Energy Product Generated
Step 3 (Isocitrate → α-KG)	Isocitrate Dehydrogenase	**1 NADH** produced + 1 CO_2 released
Step 4 (α-KG → Succinyl-CoA)	α-Ketoglutarate Dehydrogenase	**1 NADH** produced + 1 CO_2 released
Step 5 (Succinyl-CoA → Succinate)	Succinyl-CoA Synthetase	**1 ATP** (via GTP substrate phosphorylation)
Step 6 (Succinate → Fumarate)	Succinate Dehydrogenase	**1 $FADH_2$** produced (Complex II)
Step 8 (Malate → Oxaloacetate)	Malate Dehydrogenase	**1 NADH** produced; regenerates acceptor

Net Stoichiometry of the Citric Acid Cycle

Product	Per Acetyl-CoA (1 Turn)	Per Glucose (2 Turns)
NADH	3 NADH	**6 NADH** (to ETC Complex I)
$FADH_2$	1 $FADH_2$	**2 $FADH_2$** (to ETC Complex II)
ATP (GTP)	1 ATP	**2 ATP** (Substrate-level)
CO_2 Waste	2 CO_2	**4 CO_2** (Exhaled)

Module 6: Stage 4 • The Electron Transport Chain & Proton Pumping

Architecture of the Respiratory Chain

The **Electron Transport Chain (ETC)** consists of four multi-protein complexes (I–IV) and mobile electron carriers embedded in the folded inner mitochondrial membrane (cristae):

- **Complex I (NADH Dehydrogenase):** Accepts 2 e^- from NADH; oxidizes it to NAD^+; pumps 4 H^+ into intermembrane space.
- **Complex II (Succinate Dehydrogenase):** Accepts 2 e^- from $FADH_2$; oxidizes it to FAD; *does not pump protons*.
- **Ubiquinone (Coenzyme Q):** Hydrophobic mobile lipid shuttle in membrane core; carries electrons from Complexes I & II to Complex III.
- **Complex III (Cytochrome bc_1 Complex):** Transfers electrons from Q to Cytochrome c; pumps 4 H^+.
- **Cytochrome c:** Mobile peripheral protein on intermembrane side; carries electrons to Complex IV.
- **Complex IV (Cytochrome c Oxidase):** Passes electrons to terminal acceptor O_2; pumps 2 H^+.

Terminal Oxygen Reduction

$$O_2 + 4\,e^- + 4\,H^+ \rightarrow 2\,H_2O$$

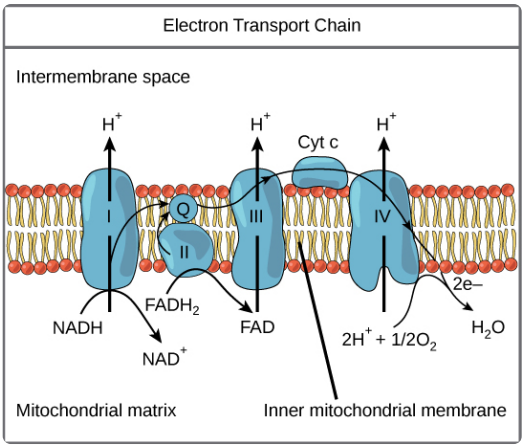


Figure 9.11: The Multi-Protein Electron Transport Chain. *Exergonic electron passage drives active proton pumping from matrix to intermembrane space.*

Cellular Poisons Inhibiting the ETC

- **Rotenone:** Blocks electron transfer from Complex I to Ubiquinone (Q). Prevents NADH utilization while allowing $FADH_2$ via Complex II.
- **Cyanide (CN^-) & Carbon Monoxide (CO):** Irreversibly bind heme iron in Complex IV, blocking oxygen reduction. Completely arrests electron flow.

Module 7: Chemiosmosis, ATP Synthase & Overall Aerobic Energy Accounting

Chemiosmosis & The Proton-Motive Force

Electron transfer along Complexes I, III, and IV generates an electrochemical gradient across the inner membrane called the **proton-motive force**:

- Protons (H^+) are concentrated in the **intermembrane space** ([H^+] high, pH ≈ 7) relative to the matrix ([H^+] low, pH ≈ 8).
- Because the lipid bilayer is impermeable to ions, protons can only re-enter the matrix by diffusing through **ATP Synthase**.

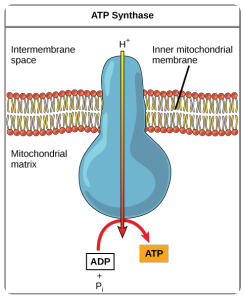


Figure 9.12: The ATP Synthase Molecular Rotary Motor. *Protons entering the F_0 channel drive internal shaft rotation, catalyzing ATP synthesis in F_1.*

Total Aerobic ATP Accounting per Glucose

Stage	Electron Coenzymes	ATP Output
1. Glycolysis	2 NADH (cytosol)	2 ATP (Substrate)
2. Pyruvate Oxidation	2 NADH (matrix)	0 ATP
3. Citric Acid Cycle	6 NADH + 2 $FADH_2$	2 ATP (Substrate)
4. Oxidative Phosphorylation	10 NADH × ~2.5 ≈ 25 ATP 2 $FADH_2$ × ~1.5 ≈ 3 ATP	~26–28 ATP (Oxidative)
TOTAL THEORETICAL YIELD	**10 NADH + 2 $FADH_2$**	**~30–32 ATP**

Why ATP Yield Is Inexact (~30–32)

1. Phosphorylation ratio is non-integer (~2.5 ATP/NADH, ~1.5 ATP/$FADH_2$).
2. Cytosolic NADH requires shuttle systems (malate-aspartate vs. glycerol-phosphate).
3. Proton-motive force also powers pyruvate transport into the matrix.

Module 8: Fermentation Pathways & Anaerobic Respiration

The Anaerobic Dilemma: Regenerating NAD^+

Without molecular oxygen (O_2), the electron transport chain stops. NADH cannot donate its electrons, causing NAD^+ levels to plummet:

- Without free NAD^+, **glycolysis shuts down** at Step 6, arresting all ATP synthesis.
- **Fermentation** is an anaerobic metabolic adaptation that transfers electrons from NADH onto organic molecules, **regenerating NAD^+** to keep glycolysis producing 2 net ATP.

Alcohol Fermentation

Occurs in yeasts (e.g., *Saccharomyces*) and many plant tissues:

1. Pyruvate (3C) is decarboxylated by pyruvate decarboxylase, releasing **CO_2** to yield acetaldehyde (2C).
2. Acetaldehyde is reduced by NADH, yielding **ethanol** and regenerating NAD^+.
3. Commercial uses: Brewing alcohol and baking bread.

Lactic Acid Fermentation

Occurs in human skeletal muscle during intense anaerobic exertion and in dairy bacteria (e.g., *Lactobacillus*):

- Pyruvate is reduced directly by NADH to form **lactate** without releasing CO_2.
- Regenerates NAD^+ immediately to sustain rapid cytosolic ATP production.
- Lactate is carried by the bloodstream to the liver, where it is converted back to pyruvate or glucose (Cori cycle).

Comparison: Respiration vs. Fermentation

Feature	Aerobic Respiration	Fermentation
Terminal Acceptor	Molecular Oxygen (O_2)	Organic molecule (Acetaldehyde/ Pyruvate)
ATP Yield / Glucose	**~30–32 ATP**	**2 ATP** (from glycolysis only)
Carbon End Products	CO_2 (fully oxidized)	Ethanol or Lactate (organic fuels)
Site of Occurrence	Cytosol + Mitochondria	Cytosol only

Module 9: Course Active Learning Practice Problems • Part I (Pathway Accounting)

Classroom Pathway Accounting (Active Learning #3, Slides 4–6)

Direct transcription of in-class board problem sets detailing product yields and destination compartments:

Pathway Stage	Direct Product Output	Cellular Destination & In-Class Rationale
Glycolysis (in Cytosol)	**2 Pyruvate**	Transported through mitochondrial carrier into matrix for Step 2.
	4 ATP (2 Net)	Substrate-level phosphorylation; immediately used for cellular work.
	2 NADH	Shuttled to mitochondrial inner membrane to feed ETC.
Pyruvate Oxidation (per Pyruvate; x2 per Glucose)	**1 Acetyl-CoA** (2 per glucose)	Stays in mitochondrial matrix for Citric Acid Cycle (Step 3).
	1 NADH (2 per glucose)	Stays in mitochondrial matrix; donates electrons directly to Complex I.
	1 CO_2 (2 per glucose)	Diffuses out of mitochondrion and cell; **exhaled!**
Citric Acid Cycle (per Acetyl-CoA; x2 per Glucose)	**3 NADH** (6 per glucose)	Stays in matrix; donates electrons to Complex I of ETC.
	1 $FADH_2$ (2 per glucose)	Stays in matrix; donates electrons to Complex II of ETC.
	1 ATP (2 per glucose)	Generated via substrate-level phosphorylation; immediately used.
	2 CO_2 (4 per glucose)	Fully oxidized carbon waste; diffused and **exhaled!**

Oxidative Phosphorylation Chemiosmosis Rules (Slide 6)

- **Active Transport:** H^+ ions are pumped actively from the **Matrix into the Intermembrane Space** against their electrochemical gradient, powered by electron transfers.
- **NADH vs. $FADH_2$ Pathway:** NADH enters at **Complex I** → Q → **Complex III** → Cyt c → **Complex IV** → O_2. $FADH_2$ enters at **Complex II** → Q → **Complex III** → Cyt c → **Complex IV** → O_2.

Module 9: Course Active Learning Practice Problems • Part II (Cellular Poisons Matrix)

Cellular Poisons Diagnostic Analysis (Active Learning #3, Slide 7)

Analyze the five experimental respiratory poisons and diagnostic symptoms evaluated in lecture:

Poison	Molecular Mechanism	Diagnostic Symptoms & Step-by-Step Rationale
Shikonin	Inhibits primary breakdown of glucose	**Extremely high intracellular glucose** accompanied by negligible ATP synthesis, as glucose cannot enter glycolysis.
Arsenic	Inhibits pyruvate dehydrogenase (pyruvate → Acetyl-CoA)	**Massive pyruvate accumulation** with normal glucose levels; Citric Acid Cycle is starved of Acetyl-CoA.
Rotenone	Blocks electron transfer from Complex I to Ubiquinone	**Elevated NADH levels**; $FADH_2$ consumption remains intact via Complex II; ATP production is moderately reduced.
DNP	Uncouples inner membrane by making it leaky to H^+	**Collapses proton gradient** (H^+ leaks into matrix); electron transport accelerates and releases heat without synthesizing ATP.
Oligomycin	Inhibits the F_0 rotor channel of ATP Synthase	**Extreme H^+ accumulation in intermembrane space** with completely arrested ATP synthesis; protons cannot re-enter matrix.

Diagnostic Decision Flowchart

- If **glucose is high and ATP is low** → **Shikonin** (glycolysis blocked).
- If **pyruvate is high and glucose is normal** → **Arsenic** (pyruvate oxidation blocked).
- If **NADH is high but $FADH_2$ is normal** → **Rotenone** (Complex I blocked).
- If **protons leak, membrane uncouples, and heat rises** → **DNP** (uncoupler).
- If **protons accumulate in IMS and cannot enter matrix** → **Oligomycin** (ATP synthase blocked).

Module 10: Foundational Practice Examination • Part I (Questions 1 – 7)

1. Where in the eukaryotic cell does glycolysis take place?

A) Mitochondrial matrix.
B) Intermembrane space.
C) Inner mitochondrial membrane (cristae).
D) Cytosol / Cytoplasm.

2. What is the net production of ATP and NADH from one molecule of glucose during glycolysis?

A) 4 ATP and 2 NADH.
B) 2 ATP and 2 NADH.
C) 2 ATP and 4 NADH.
D) 32 ATP and 10 NADH.

3. If a cell has glucose and all glycolytic enzymes present with ample oxygen, but zero ATP, will glycolysis occur?

A) Yes, because glycolysis is an exergonic pathway that releases energy.
B) No, because the energy investment phase requires 2 ATP to phosphorylate glucose.
C) Yes, because hexokinase can use inorganic phosphate instead of ATP.
D) No, because glycolysis requires carbon dioxide as an activator.

4. What is the primary function of coenzymes NAD^+ and FAD in cellular respiration?

A) To directly hydrolyze glucose into pyruvate.
B) To act as mobile electron carriers, accepting high-energy electrons and delivering them to the ETC.
C) To pump protons across the outer mitochondrial membrane.
D) To catalyze substrate-level phosphorylation in the cytosol.

5. During pyruvate oxidation, what happens to each three-carbon pyruvate molecule?

A) It is reduced to lactate in the cytosol.
B) It is decarboxylated to form a 2C acetyl group, oxidized (reducing NAD^+ to NADH), and attached to CoA.
C) It condenses directly with oxaloacetate in the intermembrane space.
D) It is phosphorylated into phosphoenolpyruvate using GTP.

6. For each molecule of glucose completely processed through the Citric Acid Cycle, what are the direct products?

A) 2 Acetyl-CoA, 2 NADH, and 2 ATP.
B) 4 CO_2, 6 NADH, 2 $FADH_2$, and 2 ATP.
C) 6 CO_2, 10 NADH, and 32 ATP.
D) 2 Pyruvate, 2 $FADH_2$, and 4 ATP.

7. What is the final electron acceptor of the electron transport chain in aerobic cellular respiration?

A) NAD^+.
B) Molecular Oxygen (O_2).
C) Cytochrome c.
D) Water (H_2O).

Module 10: Foundational Practice Examination • Part II (Questions 8 – 15)

8. How does oxidative phosphorylation differ fundamentally from substrate-level phosphorylation?

A) Substrate-level uses kinase enzymes to transfer phosphate from an organic molecule; oxidative uses a proton gradient and ATP synthase.
B) Substrate-level produces 90% of cellular ATP; oxidative produces only 10%.
C) Substrate-level requires oxygen; oxidative is entirely anaerobic.
D) Substrate-level occurs exclusively in mitochondria; oxidative occurs in the cytosol.

9. Protons (H^+) are pumped across the inner mitochondrial membrane by Complexes I, III, and IV. In what direction does this pumping occur?

A) From the intermembrane space into the matrix.
B) From the mitochondrial matrix into the intermembrane space.
C) From the cytosol into the intermembrane space.
D) From the matrix into the cytosol.

10. What directly provides the immediate energy to drive rotational catalysis by ATP synthase?

A) Direct transfer of electrons from NADH.
B) The exergonic flow of H^+ ions down their electrochemical gradient from the intermembrane space into the matrix.
C) The splitting of glucose into triose sugars.
D) Hydrolysis of GTP in the matrix.

11. In human skeletal muscle cells operating under severe oxygen deficit, what is the primary purpose of lactic acid fermentation?

A) To generate large quantities of ATP through chemiosmosis.
B) To oxidize NADH back to NAD^+, allowing glycolysis to continue producing ATP.
C) To synthesize glycogen directly from pyruvate.
D) To eliminate carbon dioxide from the muscle tissue.

12. A patient is exposed to the chemical poison Arsenic. Which of the following diagnostic metabolic profiles is observed?

A) Glucose accumulation with zero pyruvate produced.
B) Massive pyruvate accumulation because pyruvate dehydrogenase is inhibited.
C) Normal ATP synthesis with collapsed proton gradient.
D) High $FADH_2$ with zero NADH.

13. The chemical 2,4-dinitrophenol (DNP) makes the inner mitochondrial membrane permeable to protons. What is the effect of DNP on cellular respiration?

A) Electron transport stops immediately, halting oxygen consumption.
B) The proton gradient collapses, abolishing ATP synthesis while electron transport and heat production accelerate.
C) ATP synthase rotates at maximum speed, producing excessive ATP.
D) Glycolysis is permanently inhibited.

14. Why does $FADH_2$ yield fewer ATP molecules than NADH during oxidative phosphorylation?

A) $FADH_2$ contains less potential energy and enters the ETC at Complex II, bypassing Complex I proton pumping.
B) $FADH_2$ cannot donate electrons to ubiquinone.
C) $FADH_2$ is trapped in the cytosol and cannot enter mitochondria.
D) $FADH_2$ only donates electrons under anaerobic conditions.

15. How many total molecules of CO_2 are released from the complete aerobic breakdown of ONE glucose molecule?

A) 2 CO_2.
B) 4 CO_2.
C) 6 CO_2 (2 from pyruvate oxidation and 4 from the Citric Acid Cycle).
D) 12 CO_2.

Module 10: Comprehensive Examination Answer Key & Explanations

Q1: D • Subcellular Site of Glycolysis
Rationale: Glycolysis occurs entirely within the cytosol / cytoplasm. Pyruvate oxidation and the Citric Acid Cycle take place in the mitochondrial matrix.

Q2: B • Net Stoichiometry of Glycolysis
Rationale: Glycolysis consumes 2 ATP during investment and produces 4 ATP and 2 NADH during payoff, yielding 2 net ATP and 2 NADH per glucose.

Q3: B • Energy Investment Requirement
Rationale: Glycolysis cannot start without 2 ATP to phosphorylate glucose into fructose-1,6-bisphosphate via hexokinase and phosphofructokinase.

Q4: B • Function of Electron Coenzymes
Rationale: NAD^+ and FAD are electron acceptors that shuttle high-energy electrons from catabolic fuels to the inner membrane electron transport chain.

Q5: B • Pyruvate Dehydrogenase Catalysis
Rationale: Pyruvate oxidation cleaves a carboxyl group ($-CO_2$), oxidizes the remaining 2C fragment (producing NADH), and attaches Coenzyme A to yield Acetyl-CoA.

Q6: B • Citric Acid Cycle Products per Glucose
Rationale: Because 1 glucose produces 2 Acetyl-CoA, two turns of the CAC generate 4 CO_2, 6 NADH, 2 $FADH_2$, and 2 ATP (via GTP).

Q7: B • Terminal Electron Acceptor
Rationale: O_2 is the final electron acceptor in aerobic respiration. It accepts 4 electrons and 4 protons at Complex IV to form 2 H_2O.

Q8: A • Phosphorylation Mechanism Distinctions
Rationale: Substrate-level phosphorylation uses soluble kinases to transfer phosphate directly from a substrate to ADP; oxidative uses chemiosmotic rotary ATP synthase.

Q9: B • Proton Pumping Direction
Rationale: Complexes I, III, and IV pump protons from the mitochondrial matrix outward into the intermembrane space, creating the proton-motive force.

Q10: B • Power Source for ATP Synthase
Rationale: Protons diffuse down their electrochemical gradient from the intermembrane space back into the matrix through the F_0 rotor of ATP synthase.

Q11: B • Purpose of Anaerobic Fermentation
Rationale: Fermentation does not synthesize ATP itself; it oxidizes NADH to regenerate NAD^+, allowing glycolysis to continue producing 2 ATP per glucose.

Q12: B • Arsenic Mechanism & Symptoms
Rationale: Arsenic inhibits pyruvate dehydrogenase. Pyruvate cannot be converted to Acetyl-CoA, resulting in massive pyruvate buildup and low ATP.

Q13: B • DNP Chemical Uncoupling
Rationale: DNP shuttles protons across the inner membrane, dissipating the proton gradient. Electron transport runs rapidly releasing heat, but ATP synthesis collapses.

Q14: A • NADH vs. $FADH_2$ Energetics
Rationale: $FADH_2$ enters the ETC at Complex II, bypassing the proton pumping of Complex I. It yields ~1.5 ATP compared to ~2.5 ATP per NADH.

Q15: C • Total Carbon Dioxide Accounting
Rationale: All 6 carbons of glucose are released as CO_2: 2 CO_2 in pyruvate oxidation (matrix) and 4 CO_2 in the Citric Acid Cycle (matrix).